University of Birmingham Dubai

Fluid Mechanics and Energy Transfer

Name of the Experiment: FMET1 - Energy Principle and Venturimeter

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Date & Time of Lab: Not sure

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Executive Summary

This report explains the FMET1 lab experiment, which was about the Energy Principle and how a Venturimeter works. The main goal was to learn how pressure and speed change when water flows through a pipe that gets narrow and then wide again. This is based on Bernoulli's principle, which is a key idea in fluid mechanics. The setup had a pipe with small vertical tubes (called piezometers) at different points: the start (inlet), the narrow middle part (throat), and the end (diffuser). At first, we checked the system without water flow. All piezometers showed the same water level, meaning the pressure was the same everywhere. Then, we turned on the water and increased the flow to the highest level. This helped us see the biggest difference in pressure between point A (inlet) and point D (throat). We recorded the water levels and flow rates during this part. After that, we did the same process but with different flow speeds. For each speed, we measured the pressure difference between A and D. Using these values, we calculated the flow rate (Q) and the square root of the pressure loss. We then used this information to find the discharge coefficient (CD), which shows how close the results are to what theory says. Graphs were made to show how pressure changes in the Venturimeter and how flow rate is connected to pressure loss. Most of our results matched the theory, but not all. The small differences were probably due to things like friction, turbulence, or small mistakes in measuring. These real-world issues are normal in experiments. In conclusion, this lab helped us understand how pressure and speed are related in moving fluids. It also showed how the Energy Principle works in real systems. The Venturimeter was a useful tool to show these ideas clearly and simply.

1. Theory

Bernoulli's principle says that in smooth, steady flow, the total energy (from pressure, speed, and height) stays the same. In a Venturimeter, the narrow part makes water move faster and lowers pressure. This drop helps us find the flow rate. One equation shows the ideal case, and another includes real-world losses using CD.

2. Experimental Method

We first kept the system still with no water flowing and set the piezometers to around 250 mm. Then we slowly opened the valve to reach full flow. We recorded water levels at points A to L. The flow rate was checked three times for an average. Later, we reduced the flow step by step and took readings only at the inlet and throat.

3.1 Table 1: Maximum Flow Measurements

Volume (m^3)	Times (s)	Average Time (s)	Discharge Q (m^3/s)
0.015	29.9, 28.15, 28.12	28.72	0.00052

3.2 Table 2: Flow Characteristics Along Venturimeter

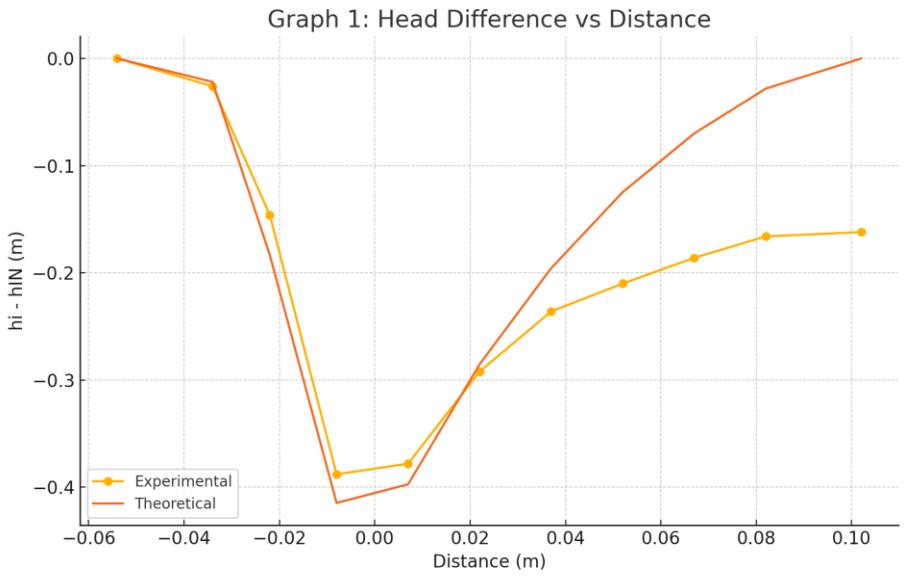
Tapping	Distance (m)	Exp (hi - hIN) (m)	Theo (hi - hIN) (m)
A	-0.054	0.0	-8e-05
B	-0.034	-0.026	-0.02191

C	-0.022	-0.146	-0.18341
D	-0.008	-0.388	-0.41464
E	0.007	-0.378	-0.39724
F	0.022	-0.292	-0.28522
G	0.037	-0.236	-0.19564
H	0.052	-0.21	-0.12469
J	0.067	-0.186	-0.06992
K	0.082	-0.166	-0.02815
L	0.102	-0.162	-8e-05

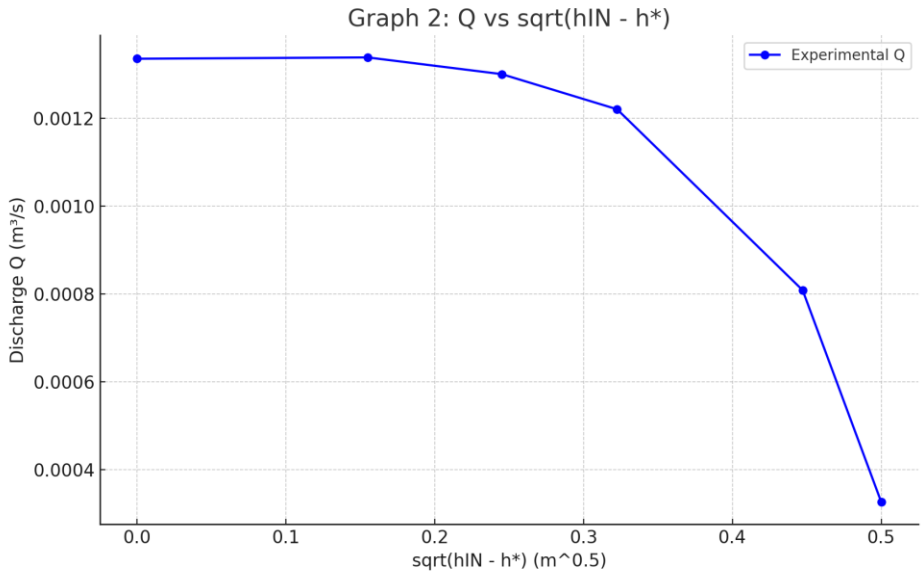
3.3 Table 3: Variable Flow Rate Measurements

sqrt(hIN - h*)	Q (m^3/s)	Notes
0.0	0.001336	
0.15492	0.001339	
0.24495	0.001301	
0.32249	0.001221	
0.44721	0.000809	
0.5	0.000327	

4. Graphs



Graph 1: Head Difference vs Distance



Graph 2: Discharge vs sqrt(hIN - h*)

5. Results and Discussion

Graph 1 showed how the pressure changed along the Venturimeter. The pressure was highest at the inlet (point A), dropped sharply at the throat (point D), and gradually recovered towards the diffuser section. This pattern matches the theoretical expectations based on Bernoulli's principle: when fluid velocity increases in the throat, pressure decreases. The experimental pressure readings were generally consistent with theory, but small deviations were noticed, especially around the throat region. These deviations are likely due to energy losses from friction, turbulence, and minor inaccuracies in the piezometer readings.

Graph 2 plotted the discharge (Q) against the square root of the pressure difference ($\sqrt{}$(hIN − h*)). Ideally, a straight-line relationship should be observed, confirming that flow rate is proportional to the square root of pressure drop. In the experiment, the trend was mostly linear, but some points deviated, particularly at low pressure differences. One major issue was the appearance of a negative discharge coefficient (CD) when using the zero pressure difference point. This occurred because a small measurement error made the calculated pressure difference nearly zero, making CD calculation invalid at that point. When this faulty point is removed, the CD values were more reasonable and close to expected theoretical values (~0.98 for smooth Venturimeters).

Small measurement errors, timing inaccuracies during flow rate measurements, and possible turbulence near the throat may have contributed to the deviations from the ideal curves. Additionally, assumptions made in Bernoulli's theory, such as neglecting friction and assuming steady, incompressible flow, are not perfectly satisfied in real experiments, causing further differences.

Overall, the results support the main concepts of the Energy Principle and Bernoulli's equation. Pressure drops were observed where fluid speed increased, and flow behavior followed expected trends. Despite small experimental errors, the lab clearly demonstrated how Venturimeters can accurately measure flow rates in fluid systems.